Assignment-4

K. R. MANGALAM UNIVERSITY

BTech (AI & ML) SOET
(School of Engineering and Technology)

Name: MOHIT GOEL

Roll No: 2301730109

Course: Computer Networks Lab

Course Code: (ENCS304)

K. R. MANGALAM UNIVERSITY, SOHNA ROAD
GURUGRAM, HARYANA

Experiment Objectives:

Configure and test a Domain Name System (DNS) Server to resolve domain names into IP addresses using Cisco Packet Tracer.

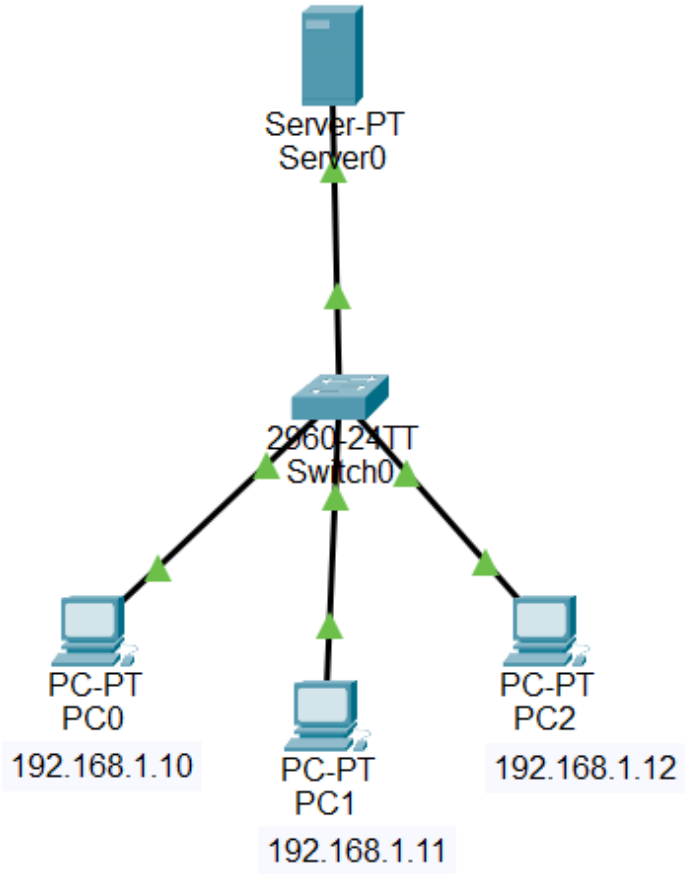
Learning Outcomes

- Understand working of DNS (name resolution process)
- Configure DNS server in Packet Tracer
- Verify DNS resolution using commands
- Analyze DNS query-response mechanism

Logical structure should include:

- **Network Devices:** Server, PCs, Switch
- **Services:** DNS Service
- **Connections:** Copper straight-through cables
- **Data Flow:** DNS query → DNS server → response

Network Topology :



Connection Tables:

Copper Straight-Through Connections

From	To	Port/Interface
PC0	Switch	fa0/1
PC1	Switch	fa0/2
Switch	Server	fa0/3

- **DNS Server Configuration**

Steps:

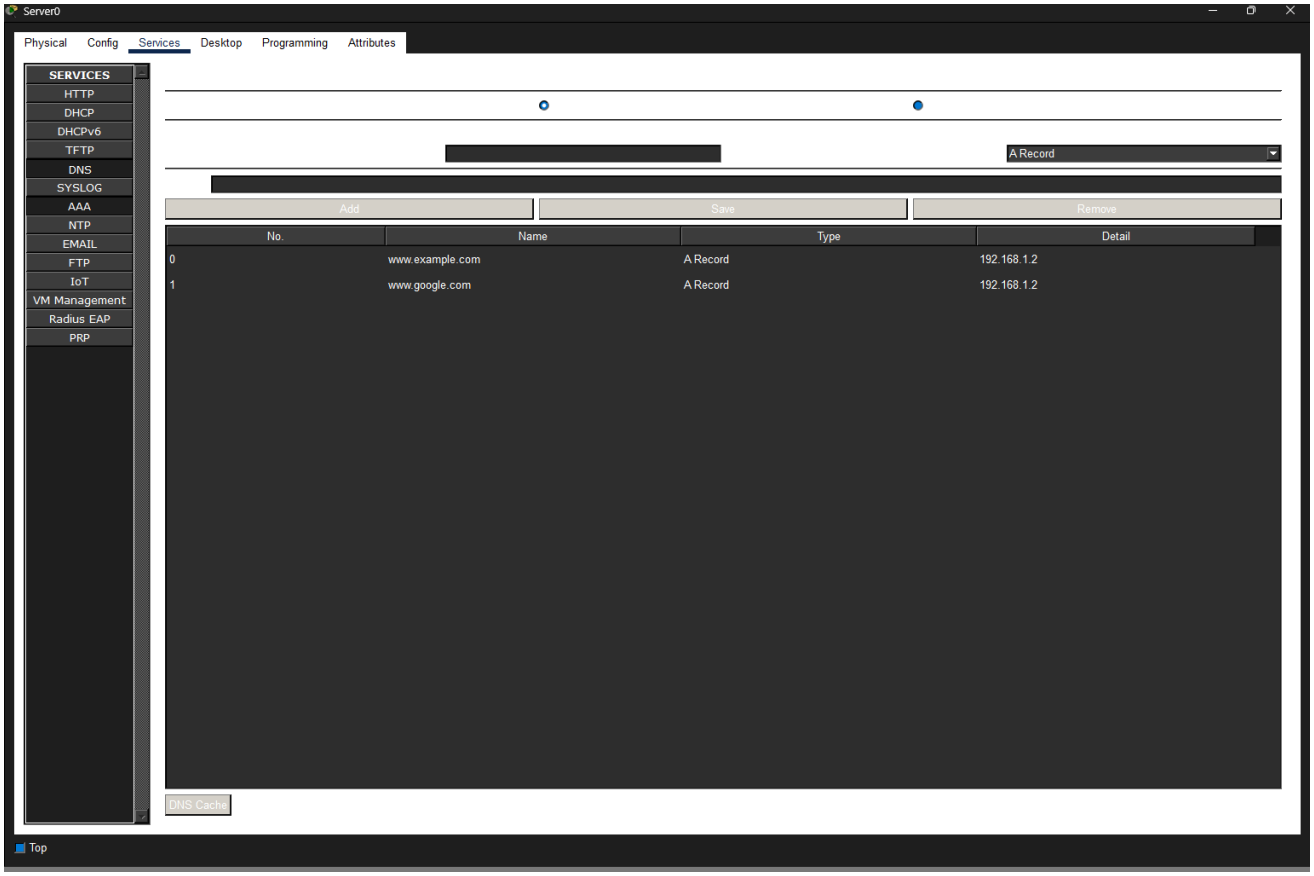
1. Open Cisco Packet Tracer
2. Add:
   - 1 Server
   - 2–3 PCs
   - 1 Switch
3. Connect using copper cables
4. Assign IP:

Device	IP Address
DNS Server	192.168.1.2
PC0	192.168.1.3
PC1	192.168.1.4

- **DNS Configuration:**
  - Go to **Server → Services → DNS**
  - Turn **DNS ON**

- **Add Records:**

Domain Name	IP Address
www.example.com	192.168.1.2
www.google.com	192.168.1.2



- **PC Configuration**
- Set IP manually
- Set **DNS Server = 192.168.1.2**

- **Working of DNS**

DNS (Domain Name System) converts domain names into IP addresses.

Process:

1. User enters domain (www.example.com)
2. Request goes to DNS server
3. DNS server checks records
4. Returns IP address
5. Communication starts

- **Simulation of Network Operation**
    1. Switched to **Simulation Mode**
    2. Selected:
        - DNS
        - ICMP

3. Sent requests from PC to server
4. Observed query-response

Output:

Ping Command

C:\>ping www.example.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TIL=128
Reply from 192.168.1.2: bytes=32 time=1ms TIL=128
Reply from 192.168.1.2: bytes=32 time=1ms TIL=128

Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = Oms, Maximum = 1ms, Average = 0ms

C:\>ping www.google.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms

NSLOOKUP
C:\>nslookup www.example.com

Server: [192.168.1.2]
Address: 192.168.1.2
Non-authoritative answer:
Name : www.example.com
Address: 192.168.1.2

C:\>nslookup www.google.com

Server: [192.168.1.2]
Address: 192.168.1.2

Non-authoritative answer:
Name: www.google.com
Address: 192.168.1.2
TraceRoute
C:\>tracert www.example.com

Tracing route to 192.168.1.2 over a maximum of 30 hops:

192.168.1.2

192.168.1.2

www.example.com
192.168.1.2

192.168.1.2

www.google.com
192.168.1.2

0 ms

Trace complete.

1

1 ms

1 ms

```
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping www.google.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>nslookup www.example.com

Server: [192.168.1.2]
Address:  192.168.1.2

Non-authoritative answer:
Name:   www.example.com
Address:   192.168.1.2

C:\>nslookup www.google.com

Server: [192.168.1.2]
Address:  192.168.1.2

Non-authoritative answer:
Name:   www.google.com
Address:  192.168.1.2

C:\>
C:\>traceroute www.example.com
Invalid Command.

C:\>tracert www.example.com

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1   0 ms      1 ms      1 ms      192.168.1.2
```